

CANDIDATE NAME _____

INDEX NUMBER _____

CG _____



SERANGOON JUNIOR COLLEGE

JC2 Preliminary Examination 2013

H2 BIOLOGY

Paper 2

9648

19 August 2013/ Monday

2 Hours

Additional materials:

Writing papers / OMS

INSTRUCTIONS TO CANDIDATES

Write your name, CG and index number in the spaces at the top of this page and on all separate writing papers used.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Section A

Answer **all** questions.

Write your answers in spaces provided on the question paper.

Section B

Answer **only one** question out of two.

Write your answers on the separate answer paper provided.

INFORMATION FOR CANDIDATES

The intended number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE	
Section A	
1	/10
2	/10
3	/10
4	/10
5	/13
6	/13
7	/10
8	/4
Section B	
9	/20
10	/20
TOTAL	/100

This question paper consists of 21 printed pages and 1 blank page

Question 1

Gene expression refers to genes being 'turned on' and producing a product. The product could be an enzyme, a structural protein, or a control molecule.

Eukaryotes can regulate their gene expression at various levels whereas prokaryotes regulate gene expression predominantly at the transcriptional level.

a) Explain how the process of transcription is regulated in eukaryotes. [2]

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b) Explain why prokaryotes regulate gene expression predominantly at the transcriptional level. [2]

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c) Post transcriptional modification in eukaryotes involves the addition of the 5' guanosine cap, splicing as well as the addition of the 3' polyA tail. Explain what will happen to the resulting protein structure if a mutation occurs at the splice site of an intron. [3]

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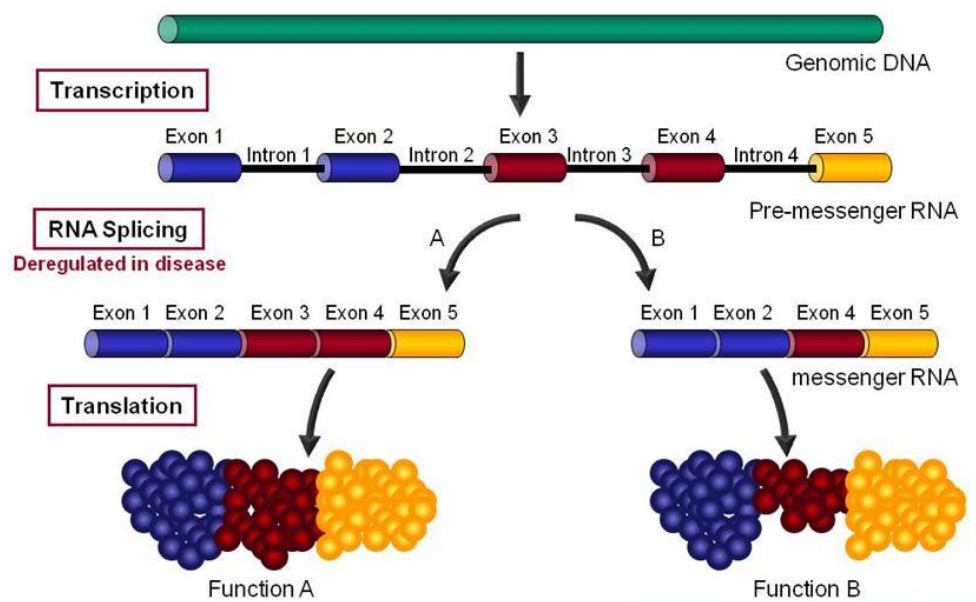


Figure 1.1

d) Name and describe the process that is shown in **Figure 1.1**. [2]

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e) Suggest why there is a need for this process. [1]

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[Total: 10 marks]

Question 2

a) Sex hormone estrogen promotes mammary epithelial cell proliferation by stimulating the expression of genes encoding cell cycle regulatory proteins. Estrogen receptors (ER α) are intracellular receptors of estrogen. Estrogen-receptor complexes bind to estrogen response elements (ERE) in enhancers. Estrogenic control of the cell cycle occurs by increasing the expression of *cyclin* and *myc C* which promote cell cycle progression.

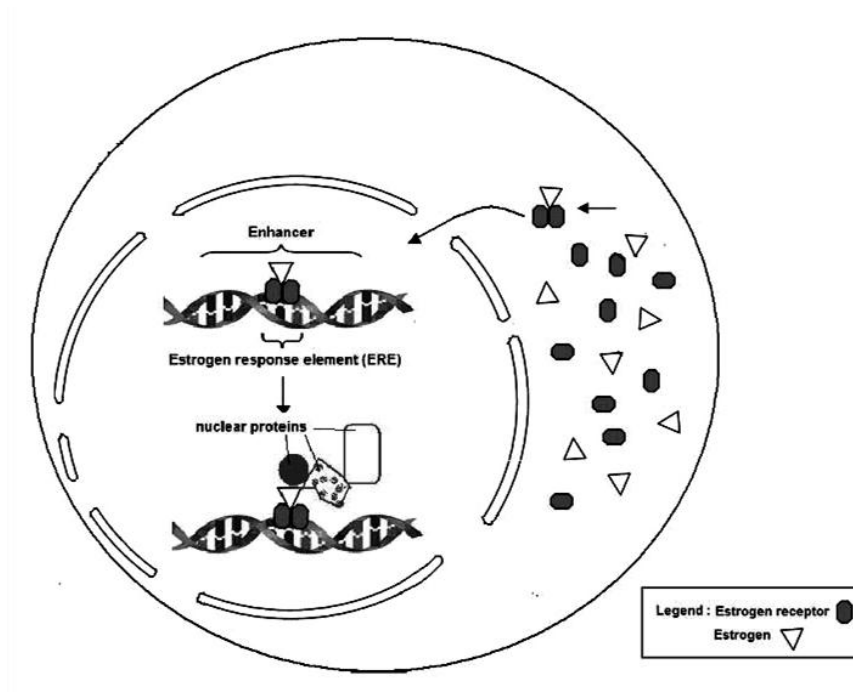


Figure 2.1

With reference to **Figure 2.1**

i) State the level at which gene expression is regulated by estrogen. [1]

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ii) Describe and explain how estrogen regulates gene expression. [4]

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b) There are two isoforms of estrogen receptors: $ER\alpha$ and $ER\beta$ encoded by different genes. Genomic DNA samples taken from women suffering from breast cancer were analysed for the gene copy number present for the $ER\alpha$ gene.

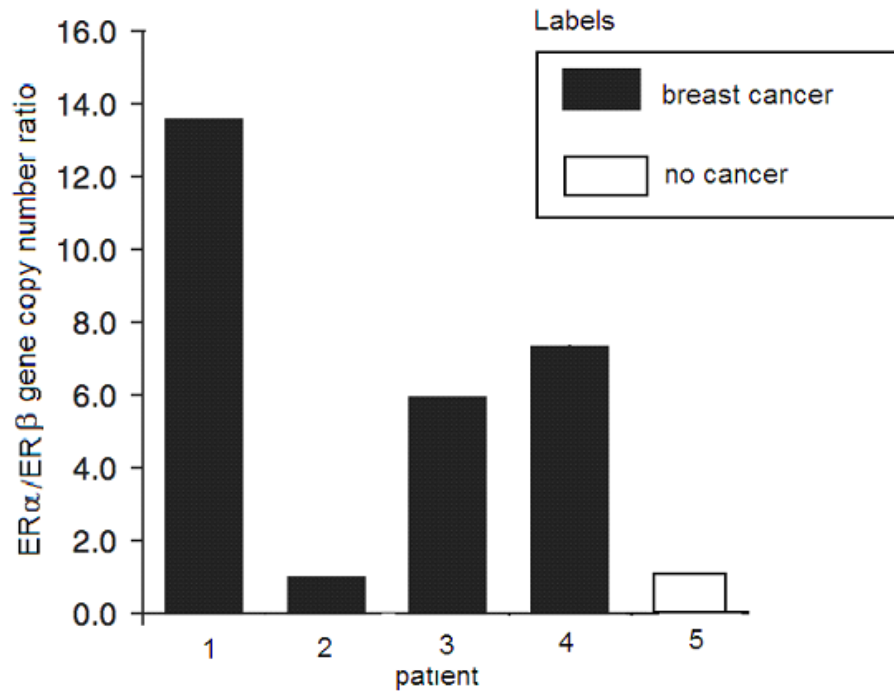


Figure 2.2

i) With reference to **Figure 2.2**, describe the effect of different $ER\alpha$ gene copy number on the women. [2]

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c) Suggest an explanation as to why patient **2** developed cancer despite a low copy number. [1]

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d) Suggest how amplification of the ER gene contributes to breast cancer. [2]

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[Total: 10 marks]

Question 3

a) A homozygous dominant black mouse was crossed with a homozygous recessive white mouse. All the F1 offspring were black. When the F1 offspring were selfed, the ratio of black to brown to white mice was 9:3:4.

Represent the above information in the form of a genetic cross. [4]

b) Explain the molecular mechanism of this cross. [3]

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c) **Table 3.1** shows the inheritance of kernel colour in wheat when 2 intermediate parents were crossed. ($AaBbCc \times AaBbCc$). The offspring contain seven different shades of kernel color based on the number of capital letters in each genotype. The higher the number of capital letters, the darker the shade of the wheat kernel.

Gametes	ABC	ABc	AbC	Abc	aBC	aBc	abC	abc
ABC	6	5	5	4	5	4	4	3
ABc	5	4	4	3	4	3	3	2
AbC	5	4	4	3	4	3	3	2
Abc	4	3	3	2	3	2	2	1
aBC	5	4	4	3	4	3	3	2
aBc	4	3	3	2	3	2	2	1
abC	4	3	3	2	3	2	2	1
abc	3	2	2	1	2	1	1	0

Table 3.1

i) Identify and state the characteristics of such an inheritance. [3]

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[Total: 10 marks]

Question 4

a) Antibiotic medications are used to kill bacteria, which can cause illness and disease. They have made a major contribution to human health. Many diseases that once killed people can now be treated effectively with antibiotics. However, some bacteria have become resistant to commonly used antibiotics.

Antibiotic resistant bacteria are bacteria that are not controlled or killed by antibiotics. Due to mutation, they are able to survive and even multiply in the presence of an antibiotic. Most infection-causing bacteria can become resistant to at least some antibiotics. Bacteria that are resistant to many antibiotics are known as multi-drug resistant organisms (MROs).

Antibiotic resistance can cause serious disease and is an important public health problem. It can be prevented by minimising unnecessary prescribing and overprescribing of antibiotics, the correct use of prescribed antibiotics, and good hygiene and infection control.

i) Explain why a mutation acquired by a bacteria will most certainly affect its phenotype. [2]

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ii) Explain how natural selection has led to the emergence of such antibiotic resistant bacteria. [4]

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b) According to the Red Queen hypothesis, sexual reproduction persists because it enables many species to rapidly evolve new genetic defenses against parasites that attempt to live off them.

Scientists have tested this idea by observing different groups of small fish *Poeciliopsis* species (Gila Topminnow) in Mexico.



Some populations of the topminnow reproduce sexually, while others practice parthenogenesis. Parthenogenesis occurs when females produce offspring without any male contribution and the female's gametes develop directly into female offspring.

Topminnows are constantly parasitized by 'black spot disease' caused by black spot worms that encyst in the skin. Parasitized topminnows rarely survive. The researchers found that identical populations of the asexually reproducing topminnows harbored many more black-spot worms than those reproducing sexually, a finding that fit the Red Queen hypothesis: the sexual topminnows could devise new defenses faster by recombination than the asexually reproducing clones.

However, it was observed that after a drought, the sexually reproducing topminnows were more heavily parasitized than the cloned topminnows. This process of whereby chance events cause the allele frequency to change unpredictably is known as genetic drift.

i) With reference to the example of topminnow, explain the concept of genetic drift and how it can alter the population's gene pool significantly. [2]

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ii) Topminnows are brought by man to a new place with totally different environmental conditions. Over time, speciation occurs and a new species of fish related to the topminnows is discovered.

Explain how speciation occurred to give rise to the new species of fish. [2]

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[Total: 10 marks]

Question 5

Figure 5.1 shows a root tip of the onion plant. The meristematic region contains cells undergoing rapid nuclear division.

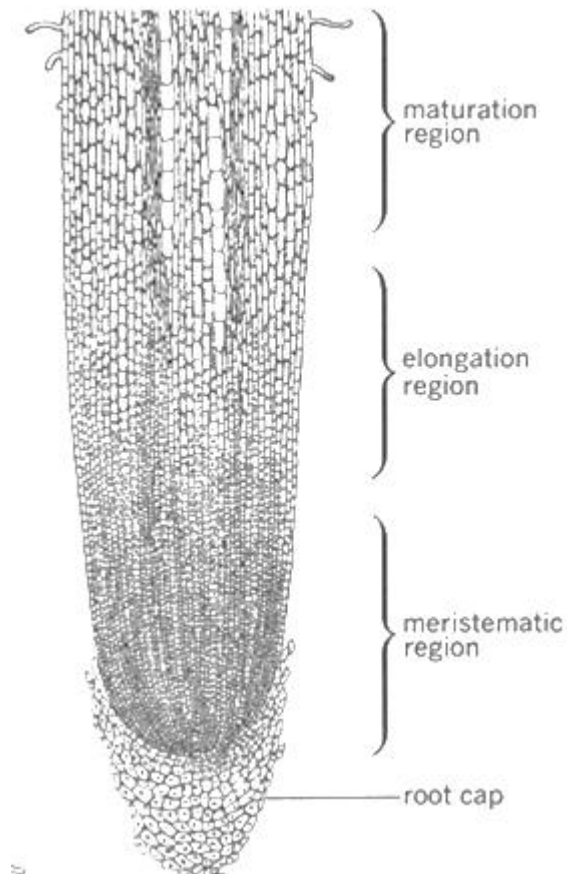


Figure 5.1

- (a) Explain the function of the type of nuclear division that is occurring in the cells found in the meristematic region in **Figure 5.1**. [3]

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(b) Describe how two distinct daughter cells are formed at the end of nuclear division. [2]

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A researcher randomly extracted two meristematic cells. He proceeded to analyse them and found them to be genetically non-identical. He concluded and wrote down in his research log book that this was due to “independent assortment of chromosomes before non-identical sister chromatids are pulled apart to opposite poles of the dividing cell”.

(c) Do you agree with the researcher’s conclusion? Explain why. [3]

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The graph in **Figure 5.2** shows the amount of DNA in the nuclei of cells taken from the meristematic region of the onion plant.

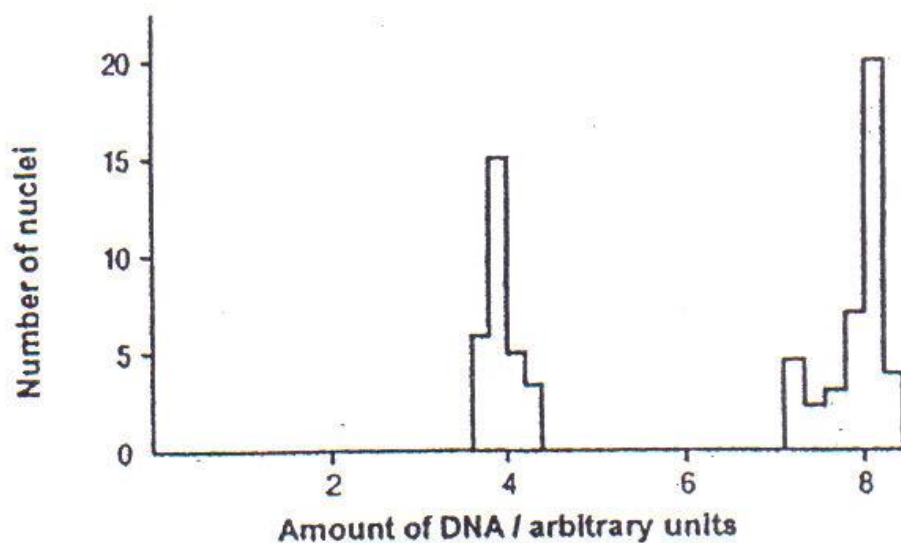


Figure 5.2

(d) With reference to **Figure 5.2**, state precisely the molecular process that accounts for the change in DNA amount in the nuclei of cells from 4au to 8 au. [1]

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(e) Compare the above process in (d) with transcription. [4]

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[Total: 13 marks]

Question 6

(a) **Figure 6.1** shows a generalized graph of the relationship between HIV copies and CD4 counts over the average course of untreated HIV infection.

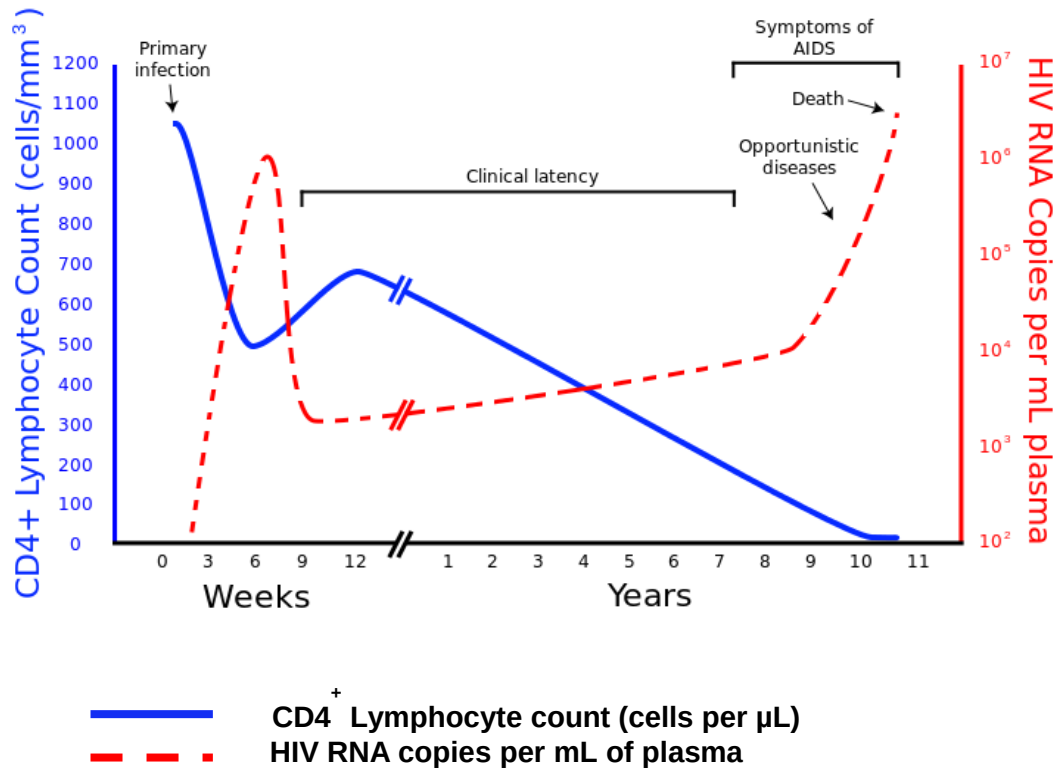


Figure 6.1

- i) Describe the relationship between the graph of CD4⁺ Lymphocyte count and the graph of HIV RNA copies. [1]

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- ii) Explain the relationship stated in (a). [4]

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- iii) As indicated in the graph above, clinical latency refers to the period where HIV infection remains undetected and patients are mostly symptom free.

Explain how the HIV virus enables itself to remain undetected during clinical latency.
[2]

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- (b) A population of *E. Coli* was grown on a culture medium containing a mixture of glucose and lactose for over 4 hours. The growth curve was plotted onto the graph in Figure 6.2 as shown.

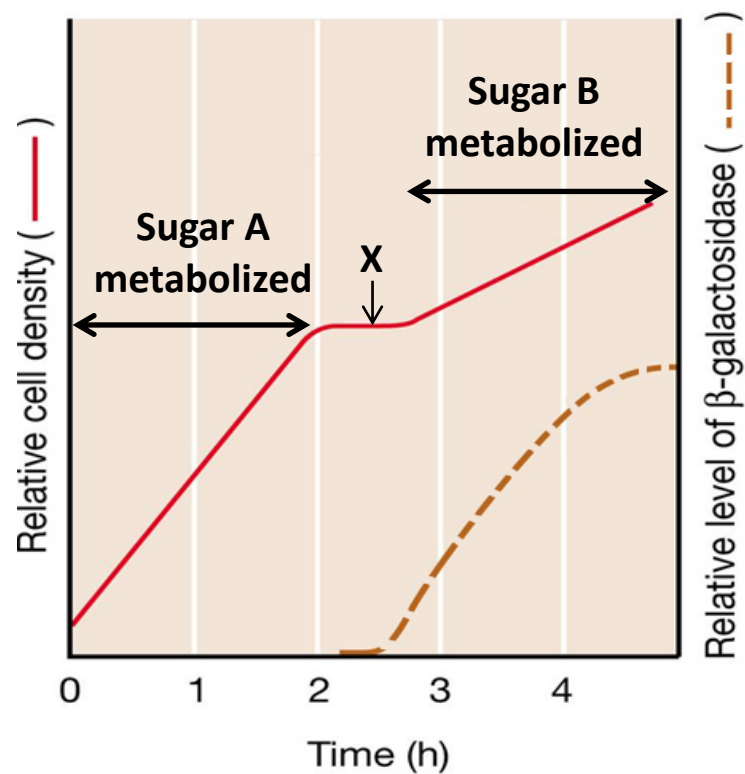


Figure 6.2

- i) Identify sugars A and B. [1]

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- ii) Outline the events that are taking place along the lac operon of *E coli* during the metabolism of sugar after 3 hours. [4]

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- iii) Suggest an explanation for the plateau marked X in **Figure 6.2**. [1]

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[Total: 13 marks]

Question 7

a) Green plants undergo photosynthesis and respiration simultaneously in the presence of light and oxygen. In an experiment, only the photosynthetic rate of two species of plants was examined and represented in **Figure 7.1**. Both plants were distinct in their leaf growth pattern. Plant A had vertical leaves while Plant B had horizontal leaves.

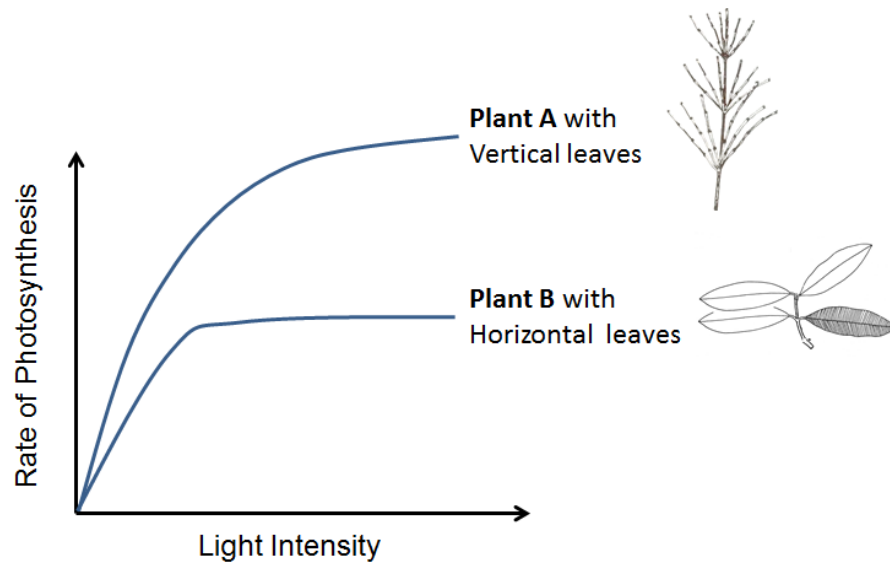


Figure 7.1

- i) Explain the difference in the shape of the graphs for Plant A and Plant B in **Figure 7.1**. [4]

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b) Plant B was extracted and placed into another environment. Conditions of the previous and new environments are tabled below in **Figure 7.2**.

Condition	Quantity	
	Previous Environment	New Environment
Amount of light	80%	0%
Amount of CO ₂	50%	0%
Amount of O ₂	21%	21%
Water availability	Yes	Yes

Figure 7.2

i) Identify the process that is occurring within Plant B. [1]

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ii) Explain the importance of oxygen in the identified process. [3]

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iii) Predict and explain briefly how the level of NADP⁺ and NAD⁺ in Plant B in this new environment compares to that in the previous environment in part a) of the question. [2]

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[Total: 10 marks]

Question 8

Radioactively-labelled amino acids were introduced into Cell A and Cell B. The muddle-headed researcher could not recall the identities of the cells, which was either the gastric epithelial cell or the cardiac muscle cell. The amount of radioactivity within three organelles (golgi apparatus, ribosomes, rough endoplasmic reticulum) in both cells were monitored over 45 min and tabled as shown in **Figure 8.1**.

	Radioactive levels / au							
	Cell A				Cell B			
	0 min	15 min	30 min	45 min	0 min	15 min	30 min	45 min
Organelle X	0	80	50	10	0	80	10	0
Organelle Y	0	0	20	15	0	60	25	5
Organelle Z	0	0	0	5	0	0	60	20

Figure 8.1

(a) Identify the organelles X, Y and Z. [1]

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(b) Identify the cell types of cells A and B and explain your answer. [3]

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[Total: 4 marks]

Section B (20 Marks)

Answer **one** question

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.

Your answers must be in continuous prose where appropriate.

Your answers must be set out in sections (a), (b), etc as indicated in the question.

Question 9

- a) Outline the differences between prokaryotic control of gene expression with the eukaryotic model. [7]
- b) Outline the significance of post-translational modifications of eukaryotic proteins. [6]
- c) Explain how the molecular structure of the axon membrane facilitates nervous transmission along the axon. [8]

Question 10

- a) Compare the Calvin cycle and the Krebs cycle. [6]
- b) Explain the importance of operons in bacteria. [4]
- c) Describe the processes through which genetic diversity can be achieved in bacteria. [10]

END OF PAPER

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